OpenClaw Model Selection — 96GB VRAM Tier

Part 7a: RTX PRO 6000 Blackwell (Native 96GB)

> **Critical constraint:** Every model listed as viable must load weights AND KV cache 100% in VRAM. No CPU offloading, no partial offloading, no hybrid RAM modes. Minimum Q4 quantization — no sub-Q4 quants are considered viable.

Executive Summary

Ninety-six gigabytes of VRAM on a single card is a qualitative threshold: the point where models designed for datacenter H100s run locally on a workstation GPU. The RTX PRO 6000 Blackwell — the only native 96GB card available — loads **GPT-oss-120B at 120–163 tokens/second**, runs **Qwen3-Next-80B-A3B with 256K context and room to spare**, and fits **Llama 3.3 70B at near-lossless Q8_0 with 32K+ context**. Three entirely new model classes unlock at this tier that were impossible at 64GB: 100B+ MoE models (GPT-oss-120B, Llama 4 Scout 109B), next-generation hybrid-attention MoE (Qwen3-Next-80B-A3B), and 70B dense models at quality-preserving Q8_0 quantization with generous context windows.

Key findings:

- **GPT-oss-120B is the headline model** — 117B MoE with only 5.1B active parameters, MXFP4 weights at ~63–65 GB, delivering 120–163 t/s on the RTX PRO 6000. It outperforms o4-mini on AIME and achieves 90% MMLU with configurable reasoning depth and native tool calling.
- **Qwen3-Next-80B-A3B is the efficiency revelation** — 80B MoE with 3B active, Q4_K_M at 48.5 GB, leaving ~47 GB for KV cache (128K+ practical context). Its hybrid Gated DeltaNet + Gated Attention architecture delivers 10× the inference throughput of Qwen3-32B on long contexts. Outperforms Gemini-2.5-Flash-Thinking on multiple benchmarks.
- **Llama 3.3 70B at Q8_0** (~60 GB) delivers near-lossless quality with 32K+ context — a transformative upgrade over the Q4_K_M version at 48GB that was limited to 8–16K context.
- **20 of 57 roles upgrade at 96GB.** The upgrades concentrate in Tiers 3–5 where the jump from 70B-Q4 to 120B-MoE or 70B-Q8 produces measurable quality gains, particularly in coding (HumanEval 88%+), orchestration, and deep reasoning.
- **Diminishing returns are real but the new MoE models change the calculus.** Without GPT-oss-120B and Qwen3-Next-80B-A3B, the 96GB tier would be a marginal upgrade. With them, it represents the first tier where a single local GPU genuinely competes with cloud API quality for production agent workloads.

---

1. RTX PRO 6000 Blackwell — The Only Native 96GB Card

The NVIDIA RTX PRO 6000 Blackwell Workstation Edition is the sole GPU offering 96GB of VRAM in a single card. No consumer GPU reaches this capacity — the RTX 5090 tops out at 32GB.

Specification	RTX PRO 6000 Blackwell
VRAM	96 GB GDDR7 with ECC
Memory interface	512-bit
Memory bandwidth	1,792 GB/s
CUDA cores	24,064
Tensor cores	5th Generation (native FP4/FP8/INT8/BF16)
PCIe interface	PCIe 5.0 x16
TDP	600W
Compute capability	12.0 (Blackwell architecture)
Price (Feb 2026)	~$7,950–8,000

Native FP4 matters at this tier. The RTX PRO 6000's Blackwell tensor cores natively accelerate MXFP4 operations — the exact format GPT-oss-120B ships in. This means the 120B model runs at hardware-native precision rather than through software dequantization, yielding the 120–163 t/s speeds benchmarked by StorageReview and Hardware Corner. No previous-generation workstation card (RTX 6000 Ada at 48GB) supports this format.

---

2. New Models That Unlock at 96GB

The jump from 64GB to 96GB adds four models that were completely inaccessible at lower tiers. Two are genuinely transformative.

2.1 Complete 96GB-Exclusive Model Roster

Model	Params	Architecture	Active Params	Quant	Weight Size	KV Room	Usable Context	Fits 96GB?
GPT-oss-120B	117B	MoE	5.1B	MXFP4 (native)	~63–65 GB	~31–33 GB	64–131K	✅ Yes

Model	Params	Architecture	Active Params	Quant	Weight Size	KV Room	Usable Context	Fits 96GB?
Llama 4 Scout	109B	MoE (16 experts)	17B	Q4_K_M	65.4 GB	~30.6 GB	32–64K	✅ Yes
Qwen3-Next-80B-A3B	80B	MoE (512 experts)	3B	Q4_K_M	48.5 GB	~47.5 GB	128K+	✅ Yes
Qwen3-Next-80B-A3B	80B	MoE (512 experts)	3B	Q6_K	65.5 GB	~30.5 GB	64–128K	✅ Yes
Qwen3-Next-80B-A3B	80B	MoE (512 experts)	3B	Q8_0	84.8 GB	~11.2 GB	16–32K	✅ Yes (tight)
Command R+ 104B	104B	Dense	104B	Q4_K_M	~62 GB	~34 GB	16–32K	✅ Yes

Mixtral 8x22B (141B MoE, Q4_K_M ~85.6 GB) technically loads but leaves only ~10.4 GB for KV cache — marginal for agentic use. Listed in the "doesn't fit" table due to insufficient context headroom.

2.2 Model Details

GPT-oss-120B — The Headline Unlock

OpenAI's first open-weight model since 2019, released August 2025 under Apache 2.0. This is a 117B-parameter MoE model with only 5.1B active parameters per token, shipped natively in MXFP4 quantization — a format the RTX PRO 6000's Blackwell tensor cores accelerate at hardware speed.

VRAM profile: The MXFP4 GGUF weights are ~63–65 GB. With Flash Attention enabled, total VRAM at 131K context drops from ~91 GB to ~67 GB — well within 96 GB with headroom. Without Flash Attention, 131K context consumes ~83 GB, still fitting but tighter.

Performance on RTX PRO 6000:

- StorageReview benchmark: **163.1 t/s** token generation (LM Studio, 500-word generation)
- Community reports: **~120 t/s** in common configurations
- vLLM with Flash Attention: **~180 t/s** at 131K context (RTX 6000 Max-Q variant)
- Prompt processing: fast enough for interactive agent use at all context lengths

Key benchmarks:

- MMLU: 90%
- AIME 2025: 98.7% (gpt-oss-20b) — the 120B exceeds o4-mini on math competitions

- HealthBench: outperforms o4-mini
- Native tool calling via Harmony 8 response format
- Configurable reasoning: low/medium/high intensity settings
- 128K native context window

Why it matters for OpenClaw: GPT-oss-120B combines frontier-class reasoning with native tool calling and configurable reasoning depth in a model that runs at 120+ t/s on a single workstation GPU. The configurable reasoning is particularly valuable for agent architectures — the orchestrator can dial up reasoning intensity for complex delegation decisions and dial it down for routine tool calls, optimizing both quality and latency.

Qwen3-Next-80B-A3B — The Efficiency Breakthrough

Released September 2025 by Alibaba, this is the first model in the Qwen3-Next series. It represents a new architectural paradigm: Hybrid Attention (Gated DeltaNet + Gated Attention) combined with ultra-sparse MoE (512 experts, 10 active + 1 shared per token = 3B active of 80B total).

The architectural innovation: Standard transformer attention scales quadratically with context length. Qwen3-Next's Gated DeltaNet layers provide linear-complexity context modeling, while periodic Gated Attention layers handle global information flow. The result: 10× inference throughput versus Qwen3-32B on contexts longer than 32K tokens, at 10% of the training cost.

VRAM profile on RTX PRO 6000:

Quantization	Weight Size	KV Room	Usable Context
Q4_K_M	48.5 GB	47.5 GB	128K+ (massive headroom)
Q5_K_M	56.7 GB	39.3 GB	96–128K
Q6_K	65.5 GB	30.5 GB	64–128K
Q8_0	84.8 GB	11.2 GB	16–32K

The Q4_K_M sweet spot is remarkable: 48.5 GB of weights leaves nearly half the 96 GB card for KV cache, enabling **128K+ context at MoE inference speeds**. The hybrid attention architecture has inherently lower KV cache overhead than standard transformers (~2 GB additional per 100K tokens), making these context estimates conservative.

Available variants:

- **Instruct** — general-purpose, non-thinking mode, tool calling support
- **Thinking** — chain-of-thought reasoning, outperforms Gemini-2.5-Flash-Thinking

Key benchmarks (Thinking variant):

Benchmark	Qwen3-Next-80B-A3B-Thinking	Qwen3-32B Thinking	Qwen3-235B Thinking	Gemini-2.5-Flash Thinking
MMLU-Pro	**82.7**	79.1	84.4	81.9
MMLU-Redux	**92.5**	90.9	93.8	92.1
GPQA	**77.2**	68.4	81.1	82.8
AIME 2025	**87.8**	72.9	92.3	72.0
LiveCodeBench v6	**68.7**	60.6	74.1	61.2
IFEval	**88.9**	85.0	87.8	89.8

The Instruct variant benchmarks:

- MMLU-Pro: 80.6
- AIME 2025: 69.5
- HumanEval: 90%+ pass@1
- Arena-Hard v2: 82.7 (higher than Qwen3-235B's 79.2)
- MultiPL-E: 87.8

Why it matters for OpenClaw: This model offers Qwen3-235B-class performance in a package that fits on a single 96GB card with massive context headroom. The MoE architecture means inference is fast (comparable to the 30B-A3B models from lower tiers), and the 256K native context enables repository-scale code understanding and full-document analysis in a single pass. The Thinking variant's AIME 87.8% and LiveCodeBench 68.7% make it a genuine competitor to the R1-Distill models for reasoning tasks, but with tool calling support that the R1 distills lack.

Llama 4 Scout 109B — Multimodal MoE

Meta's Llama 4 Scout released April 2025, featuring 109B total parameters across 16 experts with 17B active per token. Multimodal (text + vision), with a theoretical 10M token context window.

VRAM profile: Q4_K_M at 65.4 GB (official ggml-org GGUF, split across 2 files). Leaves ~30.6 GB for KV cache — sufficient for 32–64K practical context.

Caveats: Community reception has been mixed. Early llama.cpp support was buggy, and benchmark performance, while strong for its generation, has been surpassed by GPT-oss-120B and Qwen3-Next-80B-A3B in most categories. The multimodal vision capability is its unique selling point — no other 100B+ model in this tier offers native image understanding.

Best role: Vision-heavy agent tasks where image understanding needs to be paired with strong language reasoning. For text-only tasks, GPT-oss-120B and Qwen3-Next-80B-A3B are superior choices.

Command R+ 104B — RAG Specialist (Legacy)

Cohere's 104B dense model (April 2024), designed specifically for enterprise RAG and tool use. At Q4_K_M (~62 GB), it fits with ~34 GB for KV cache.

Caveats: This is the oldest model in the 96GB roster. Licensed CC-BY-NC 4.0 (non-commercial default, commercial license available from Cohere). It has been surpassed by newer models on most benchmarks, but retains niche value for its 10-language multilingual RAG optimization and sophisticated citation generation. Its tool-use capabilities were strong for its generation but don't match GPT-oss-120B's Harmony 8 format or Qwen3-Next's Qwen-Agent integration.

Best role: Multilingual RAG pipelines requiring citation tracking. For most other tasks, newer models are preferable.

2.3 Key Benchmarks — 96GB Models vs 48GB Models

Benchmark	Qwen3-32B (32GB best)	Llama 3.3 70B Q4 (48GB)	GPT-oss-120B	Qwen3-Next-80B (Thinking)	Jump from 48GB
MMLU	~81%	86.0%	**90%**	92.5 (Redux)	+4–6%
HumanEval	~78%	88.4%	~90%+	90%+	+2–5%
AIME 2025	~35%	~20%	98.7% (20B)	**87.8%**	+18–68%
LiveCodeBench v6	~55%	~60%	—	**68.7%**	+9–13%
IFEval	~85%	92.1%	—	**88.9%**	~same
GPQA	~50%	~53%	—	**77.2%**	+24%

The reasoning benchmarks show the most dramatic jumps. Qwen3-Next-80B-A3B-Thinking at 87.8% AIME 2025 is in a different league from any model available at 48GB or 64GB.

---

3. 48GB/64GB-Tier Models That Improve at 96GB

The extra VRAM budget transforms existing models through higher quantization and expanded context windows.

3.1 Quantization Upgrades

Model	48GB Tier (quant / ctx)	96GB Tier (quant / ctx)	What Improves
Llama 3.3 70B	Q4_K_M (42.5 GB) / 8–16K	**Q8_0 (~60 GB)** / 32–48K	Near-lossless quality + 3× context
R1-Distill-Llama-70B	Q4_K_M (42.5 GB) / 8–16K	**Q8_0 (~60 GB)** / 32–48K	Critical: reasoning chains preserve fidelity
Nemotron-Super-49B	Q4_K_M (30.2 GB) / 32–64K	**Q8_0 (53 GB)** / 64–128K	Near-lossless + extended context
Qwen3-30B-A3B (MoE)	Q8_0 (32.5 GB) / 160K+	Q8_0 (32.5 GB) / **160K+ with massive KV room**	Already Q8 at 48GB; now with ~63 GB KV headroom
Qwen3-Coder-30B (MoE)	Q8_0 (32.5 GB) / 160K+	Q8_0 (32.5 GB) / **160K+ with massive KV room**	Same as above
Qwen3-32B	Q8_0 (34.8 GB) / 48–96K	**BF16 (~64 GB)** / 32–48K	True lossless precision (but less context)

3.2 The Q8_0 70B Breakthrough

The single most impactful upgrade at 96GB for existing models: **Llama 3.3 70B at Q8_0** (~60 GB) with ~36 GB for KV cache. Q8_0 adds only +0.001 perplexity over FP16 — effectively lossless. At Q4_K_M (the 48GB format), the perplexity penalty is +0.054. For agent roles where output precision matters — legal reasoning, code generation, orchestration decisions — this quality delta is measurable and meaningful.

The context window roughly triples: from 8–16K at Q4_K_M to 32–48K at Q8_0 on 96GB. This transforms Llama 3.3 70B from a tight-context specialist into a full-featured agent model.

R1-Distill-Llama-70B at Q8_0 sees an even larger quality impact. Chain-of-thought reasoning is particularly sensitive to quantization — small precision losses compound across long reasoning traces. Q8_0 preserves reasoning chain fidelity that Q4_K_M noticeably degrades on the hardest mathematical problems.

---

4. Inference Speed on RTX PRO 6000

All models load on a single card — no pipeline parallelism needed. This eliminates the speed penalties that plagued multi-GPU tiers. Single-card inference with 1,792 GB/s bandwidth and native FP4 tensor core support delivers excellent throughput.

4.1 Per-Model Speed Estimates

Model	Quant	Weight Size	Architecture	Est. Token Gen	Est. Prompt Processing	Notes
GPT-oss-120B	MXFP4	~63–65 GB	MoE (5.1B active)	**120–163 t/s**	~1,000+ t/s	Benchmarked by StorageReview, Hardware Corner
Qwen3-Next-80B-A3B	Q4_K_M	48.5 GB	MoE (3B active)	**~100–140 t/s**	~800+ t/s	Est. from MoE activation ratio + bandwidth
Qwen3-Next-80B-A3B	Q6_K	65.5 GB	MoE (3B active)	**~80–110 t/s**	~700+ t/s	Slightly slower from larger weights
Qwen3-Next-80B-A3B	Q8_0	84.8 GB	MoE (3B active)	**~60–80 t/s**	~500+ t/s	Near-lossless quality trade
Llama 4 Scout 109B	Q4_K_M	65.4 GB	MoE (17B active)	**~40–60 t/s**	~500+ t/s	Higher active params = slower than GPT-oss
Llama 3.3 70B	Q8_0	~60 GB	Dense	**~25–30 t/s**	~400+ t/s	Dense model, all 70B params active
R1-Distill-70B	Q8_0	~60 GB	Dense	**~25–30 t/s**	~400+ t/s	Same architecture as Llama 3.3 70B
Nemotron-Super-49B	Q8_0	53 GB	Dense (pruned)	**~30–35 t/s**	~500+ t/s	Smaller dense = faster than 70B
Command R+ 104B	Q4_K_M	~62 GB	Dense	**~20–25 t/s**	~300+ t/s	104B dense, all params active
Qwen3-30B-A3B	Q8_0	32.5 GB	MoE (3.3B active)	**~140–180 t/s**	~1,200+ t/s	Fastest model at tier, huge KV room

4.2 Speed Tiers for Agent Role Assignment

Ultra-fast (100+ t/s): GPT-oss-120B, Qwen3-Next-80B Q4_K_M, Qwen3-30B-A3B Q8_0 — suitable for all interactive agent roles including orchestrator, real-time coding, and rapid tool-calling loops.

Fast (40–80 t/s): Qwen3-Next-80B Q6_K/Q8_0, Llama 4 Scout — suitable for interactive roles with minor latency tolerance.

Moderate (25–35 t/s): Llama 3.3 70B Q8_0, R1-Distill-70B Q8_0, Nemotron-Super-49B Q8_0 — suitable for semi-interactive roles where quality precision matters more than speed.

Slow (20–25 t/s): Command R+ 104B — background tasks only.

---

5. Inference Tools

The RTX PRO 6000 runs all major inference frameworks as a single-card deployment. No multi-GPU configuration needed.

5.1 Tool Compatibility

Tool	Single-GPU Support	Native FP4	Best For
llama.cpp	✅ Full	✅ MXFP4 via Blackwell kernels	GGUF models, quick testing, daily use
Ollama	✅ Full	✅ Via llama.cpp backend	Zero-config deployment, OpenClaw integration
vLLM	✅ Full	✅ FP8 + MXFP4 support	Production serving, continuous batching
LM Studio	✅ Full	✅ Via llama.cpp backend	Desktop GUI, easy model switching
ExLlamaV3	✅ Full	Partial	Maximum quantization quality (EXL3)
SGLang	✅ Full	✅	MoE-optimized expert parallelism
TensorRT-LLM	✅ Full	✅ Native Blackwell support	Maximum throughput (compiled engines)

5.2 Tool Recommendations

For OpenClaw daily deployment → Ollama or llama.cpp server. First-class OpenClaw integration via `OLLAMA_API_KEY="ollama-local"`. Single command to run any model: `ollama run gpt-oss:120b`. The simplicity of single-card deployment means no multi-GPU configuration headaches.

For maximum GPT-oss-120B performance → vLLM or TensorRT-LLM. The 180 t/s achieved via vLLM with Flash Attention significantly exceeds the ~120 t/s from llama.cpp. TensorRT-LLM's MoE backend (supported on Blackwell) offers even higher throughput for production workloads.

For Qwen3-Next-80B-A3B → SGLang. Expert parallelism in SGLang is well-optimized for high-sparsity MoE models. With 512 experts and only 10 active, SGLang's routing can be particularly efficient. vLLM also

supports Qwen3-Next natively (v0.10.2+) with Multi-Token Prediction for additional speed.

5.3 Dual-Model Serving on Single Card

With 96GB, simultaneous dual-model serving becomes practical for many model combinations:

Model A	Size	Model B	Size	Total	KV Room	Use Case
GPT-oss-120B	~65 GB	Qwen3-8B Q4	~5.5 GB	~70.5 GB	~25.5 GB	Frontier workhorse + fast router
Qwen3-Next-80B Q4_K_M	48.5 GB	Qwen3-14B Q4	~9 GB	~57.5 GB	~38.5 GB	Next-gen MoE + general mid-range
Nemotron-Super-49B Q8_0	53 GB	Qwen3-8B Q4	~5.5 GB	~58.5 GB	~37.5 GB	Reasoning workhorse + router
Qwen3-30B-A3B Q8_0	32.5 GB	Llama 3.3 70B Q4_K_M	42.5 GB	~75 GB	~21 GB	Fast MoE + frontier dense

The first configuration — GPT-oss-120B + Qwen3-8B router — is particularly powerful for OpenClaw. The 8B model handles all Tier 1 routing and classification at ~186 t/s while the 120B handles all complex tasks at ~120 t/s, with zero model swap latency between them.

---

6. Complete Agent Role Mapping — All 57 Roles at 96GB

6.1 Tier 1 — Simple Roles (8 roles) — No Upgrades

Every Tier 1 role was well-served at 32GB. The 96GB VRAM is wasted on these tasks.

Role	Model	Quant	Speed	Upgrade?
Router/triage	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
Input validator	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
Heartbeat/health	Qwen3-0.6B	Q8_0	~500+ t/s	➡ No
Notification/alert	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
Sentiment analysis	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
FAQ generation	Qwen3-14B	Q4_K_M	~124 t/s	➡ No
Translation	Qwen3-14B	Q4_K_M	~124 t/s	➡ No
Calendar/scheduling	Qwen3-8B	Q4_K_M	~186 t/s	➡ No

6.2 Tier 2 — Moderate Roles (25 roles) — 4 Upgrades

Role	96GB Model	Quant	Speed	Upgrade?	Why
Research/web search	**Qwen3-Next-80B-A3B**	Q4_K_M	~120 t/s	⬆ YES	128K+ context + AIME 69.5% reasoning depth
Content writing	**Qwen3-Next-80B-A3B**	Q4_K_M	~120 t/s	⬆ YES	Arena-Hard 82.7 surpasses 235B; MoE speed
Editing	Qwen3-32B	Q8_0	~40 t/s	➡ Marginal	Q8_0 already available at 48GB
Email drafting	Qwen3-32B	Q4_K_M	~50 t/s	➡ No	32B sufficient
Document summarization	**Qwen3-Next-80B-A3B**	Q4_K_M	~120 t/s	⬆ YES	128K context for entire documents in one pass
Meeting notes	Qwen3-30B-A3B	Q8_0	~150 t/s	➡ No	Already excellent at 48GB
Social media	Qwen3-14B	Q4_K_M	~124 t/s	➡ No	Short-form, 14B sufficient
News aggregation	Qwen3-14B	Q4_K_M	~124 t/s	➡ No	Classification + summary

Role	96GB Model	Quant	Speed	Upgrade?	Why
Shopping/price comparison	Qwen3-8B	Q4_K_M	~186 t/s	➡ No	Structured extraction
Memory/knowledge mgmt	Qwen3-30B-A3B	Q8_0	~150 t/s	➡ No	Already excellent at 48GB
RAG/retrieval	**Qwen3-Next-80B-A3B**	Q4_K_M	~120 t/s	⬆ YES	128K+ context = massive retrieval windows
Data analysis	Nemotron-Super-49B	Q8_0	~32 t/s	➡ Marginal	Q8_0 improves numerical precision
Website scraping	Qwen3-8B	Q4_K_M	~186 t/s	➡ No	Parsing, not reasoning
Image description	Gemma 3 27B	Q8_0	~40 t/s	➡ No	Already Q8_0 at 48GB
Customer support	Qwen3-32B	Q4_K_M	~50 t/s	➡ No	32B sufficient
Lead scoring	Qwen3-14B	Q4_K_M	~124 t/s	➡ No	Classification task
Project summarization	Qwen3-32B	Q8_0	~40 t/s	➡ No	Q8_0 already at 48GB
Transaction/approval	Qwen3-8B	Q4_K_M	~186 t/s	➡ No	Rule-heavy
Home automation	Qwen3-8B	Q4_K_M	~186 t/s	➡ No	Command parsing
Fitness tracking	Qwen3-8B	Q4_K_M	~186 t/s	➡ No	Data logging
Recipe/cooking	Qwen3-14B	Q4_K_M	~124 t/s	➡ No	Knowledge retrieval
Personal finance	Qwen3-14B	Q4_K_M	~124 t/s	➡ No	14B handles well
SEO optimization	Qwen3-14B	Q4_K_M	~124 t/s	➡ No	Content analysis
Landing page generation	Qwen3-32B	Q8_0	~40 t/s	➡ No	Already good at 48GB

Role	96GB Model	Quant	Speed	Upgrade?	Why
Travel planning	Qwen3-14B	Q4_K_M	~124 t/s	➡ No	14B sufficient

Tier 2 summary: 4 upgrades, all driven by Qwen3-Next-80B-A3B's combination of 128K+ context and MoE speed. The upgrades target document-heavy and research roles where context length is the binding constraint.

6.3 Tier 3 — Complex Roles (11 roles) — 7 Upgrades

Role	96GB Model	Quant	Speed	Upgrade?	Why
Code generation	**GPT-oss-120B**	MXFP4	~130 t/s	⬆ YES	120B MoE with tool calling, outperforms Llama 70B
Code review	**GPT-oss-120B**	MXFP4	~130 t/s	⬆ YES	Broader code understanding at 120B
QA/test writing	**GPT-oss-120B**	MXFP4	~130 t/s	⬆ YES	Better edge-case generation
Task planning	**GPT-oss-120B**	MXFP4	~130 t/s	⬆ YES	Configurable reasoning depth for planning
Fact-checking	**Llama 3.3 70B**	Q8_0	~28 t/s	⬆ YES	Q8_0 near-lossless reduces hallucination
Critic/review	**GPT-oss-120B**	MXFP4	~130 t/s	⬆ YES	Nuanced evaluation benefits from 120B scale
Market research	**Qwen3-Next-80B-A3B**	Q4_K_M	~120 t/s	⬆ YES	128K context for multi-source synthesis
Synthesizer/aggregator	Qwen3-30B-A3B	Q8_0	~150 t/s	➡ No	160K ctx + speed still optimal
Curriculum design	Qwen3-32B	Q8_0	~40 t/s	➡ No	Already good at 48GB
Prototype generation	GPT-oss-120B	MXFP4	~130 t/s	➡ Marginal	Coding already served by Llama 70B
DevOps	Qwen3-32B	Q8_0	~40 t/s	➡ No	Config/IaC generation, 32B sufficient

Tier 3 summary: 7 upgrades. GPT-oss-120B dominates coding and planning roles at 130 t/s — it's faster than any dense 70B model while delivering stronger results. Qwen3-Next-80B-A3B handles long-context synthesis.

Llama 3.3 70B at Q8_0 provides near-lossless fact-checking.

6.4 Tier 4 — Reasoning Roles (3 roles) — ALL Upgraded

Role	96GB Model	Quant	Speed	Upgrade?	Why
Math/logic reasoning	**Qwen3-Next-80B-A3B-Thinking**	Q4_K_M	~120 t/s	⬆⬆ MAJOR	AIME 87.8% vs R1-Distill-70B's 70%. Game-changing.
STEM analysis	**Qwen3-Next-80B-A3B-Thinking**	Q4_K_M	~120 t/s	⬆⬆ MAJOR	GPQA 77.2% vs R1-Distill-70B's 65.2%
Algorithm exploration	**Qwen3-Next-80B-A3B-Thinking**	Q6_K	~90 t/s	⬆⬆ MAJOR	LiveCodeBench 68.7%, CFEval 2071

Tier 4 represents the most dramatic upgrade at 96GB. Qwen3-Next-80B-A3B-Thinking's AIME 87.8% demolishes every model available at 48GB or 64GB. The R1-Distill-70B that was the 48GB-tier reasoning champion (AIME 70%) is now decisively surpassed — and the new model runs at MoE speeds (100+ t/s) rather than dense 70B speeds (~25 t/s). This is a fundamental shift: frontier-class mathematical reasoning at interactive speeds on a single GPU.

Alt strategy for maximum math quality: Load R1-Distill-70B at Q8_0 (~60 GB) when reasoning chain fidelity at the 70B-class is preferred over raw benchmark scores. The Q8_0 quantization preserves chain-of-thought precision that Q4 noticeably degrades.

6.5 Tier 5 — Frontier Roles (10 roles) — 6 Upgrades

Role	96GB Model	Quant	Speed	Cloud still better?	Upgrade?
Orchestrator/manager	**GPT-oss-120B**	MXFP4	~130 t/s	Closing gap	⬆ YES
Software architect	**GPT-oss-120B**	MXFP4	~130 t/s	Narrowing	⬆ YES
Complex debugger	**GPT-oss-120B**	MXFP4	~130 t/s	Narrowing	⬆ YES
Legal document review	**Llama 3.3 70B**	Q8_0	~28 t/s	Yes, still	⬆ YES
Medical/health analysis	Nemotron-Super-49B	Q8_0	~32 t/s	Yes, always verify	➡ Marginal
Financial analysis	Nemotron-Super-49B	Q8_0	~32 t/s	Yes, for complex	➡ Marginal
Security analyst	**GPT-oss-120B**	MXFP4	~130 t/s	Narrowing	⬆ YES
SRE/incident response	**GPT-oss-120B**	MXFP4	~130 t/s	Partially	⬆ YES

Role	96GB Model	Quant	Speed	Cloud still better?	Upgrade?
Book writing	Qwen3-Next-80B-A3B	Q4_K_M	~120 t/s	For style/coherence	➡ Marginal
Compliance/regulatory	Llama 3.3 70B	Q8_0	~28 t/s	Yes, strongly	➡ Marginal

The orchestrator role gets its most significant upgrade yet. GPT-oss-120B's combination of native tool calling (Harmony 8 format), configurable reasoning intensity, and 90% MMLU knowledge breadth makes it the strongest local orchestrator available. The configurable reasoning is specifically valuable: dial to "low" for routine routing decisions (faster), "high" for complex multi-agent coordination (slower but more reliable). At 120+ t/s, even the high-reasoning mode is interactive.

6.6 Upgrade Summary — All 57 Roles

Tier	Total Roles	Upgraded at 96GB	Unchanged	Key Driver
Tier 1 (simple)	8	0	8	8–14B already sufficient
Tier 2 (moderate)	25	4	21	Qwen3-Next-80B 128K+ context
Tier 3 (complex)	11	7	4	GPT-oss-120B speed + quality
Tier 4 (reasoning)	3	3	0	Qwen3-Next-80B-Thinking AIME 87.8%
Tier 5 (frontier)	10	6	4	GPT-oss-120B orchestration + tool calling
Total	**57**	**20**	**37**	

6.7 Model Usage Across All Roles

Model	Roles Served	Tier Coverage	Key Advantage
GPT-oss-120B MXFP4	11 roles	T3–T5	Frontier workhorse: 120+ t/s, 90% MMLU, configurable reasoning, native tool calling. Orchestrator, coding, debugging, security, SRE, planning, architecture
Qwen3-Next-80B-A3B Q4_K_M (Instruct)	5 roles	T2–T3	Context king: 128K+ at MoE speed. Research, content writing, document summary, RAG, market research
Qwen3-Next-80B-A3B Q4_K_M (Thinking)	3 roles	T4	Reasoning champion: AIME 87.8%, LiveCodeBench 68.7%. Math, STEM, algorithms
Llama 3.3 70B Q8_0	2 roles	T3–T5	Near-lossless dense: fact-checking, legal review

Model	Roles Served	Tier Coverage	Key Advantage
Nemotron-Super-49B Q8_0	2 roles	T5	Balanced quality: medical analysis, financial analysis
Qwen3-30B-A3B Q8_0 (MoE)	2 roles	T2–T3	Speed + context: synthesis, meeting notes
Qwen3-32B Q8_0	6 roles	T2–T3	Quality general: editing, email, DevOps, curriculum, landing pages, customer support
Qwen3-14B Q4_K_M	8 roles	T1–T2	Mid-range: social media, news, finance, SEO, travel, recipes
Qwen3-8B Q4_K_M	8 roles	T1	Fast lightweight: router, validator, alerts, sentiment, scheduling, scraping
Qwen3-0.6B Q8_0	1 role	T1	Heartbeat monitor

7. Deployment Architectures at 96GB

7.1 Architecture A — Single Frontier Model

Load one large model and swap as needed.

Default: GPT-oss-120B MXFP4 (~65 GB) with ~31 GB for KV cache. Covers orchestrator, coding, planning, and all Tier 3–5 roles. Swap to Qwen3-Next-80B-A3B-Thinking for deep reasoning tasks.

Swap time: ~15–25 seconds for 65 GB models from NVMe SSD.

7.2 Architecture B — Frontier + Router (Recommended for OpenClaw)

Load two models simultaneously:

- **GPT-oss-120B** MXFP4 (~65 GB) — handles all complex tasks
- **Qwen3-8B** Q4_K_M (~5.5 GB) — handles routing, triage, notifications, heartbeat

Total: ~70.5 GB. Leaves ~25.5 GB for KV cache across both models.

Zero model swap latency for the vast majority of agent interactions. Only swap to specialized models (Qwen3-Next-80B Thinking, Llama 3.3 70B Q8_0) for infrequent specialized tasks.

7.3 Architecture C — Context-Optimized

Load Qwen3-Next-80B-A3B Q4_K_M (48.5 GB) with ~47.5 GB for KV cache. This configuration provides

128K+ context at MoE speed — ideal for document-heavy workflows, RAG pipelines, and research agent workloads that need to ingest entire documents or codebases.

Optionally co-load Qwen3-14B Q4_K_M (~9 GB) for mid-range tasks, totaling ~57.5 GB with ~38.5 GB for KV cache.

7.4 Architecture D — Maximum Quality

Load Llama 3.3 70B Q8_0 (~60 GB) for near-lossless dense model quality. Leaves ~36 GB for KV cache (32–48K context). Best for roles where quantization-induced quality loss matters: legal analysis, compliance, fact-checking, and any domain requiring maximum factual precision.

7.5 Recommended Architecture by Workload

Workload	Best Architecture	Models	Why
OpenClaw multi-agent pipeline	**B — Frontier + Router**	GPT-oss-120B + Qwen3-8B	Zero-swap for 80%+ of tasks, 120+ t/s
Heavy coding sessions	A — Single frontier	GPT-oss-120B all day	Best tool calling + coding quality
Deep reasoning tasks	A — Swap to specialist	Qwen3-Next-80B-Thinking	AIME 87.8% at MoE speed
Document-heavy analysis	C — Context-optimized	Qwen3-Next-80B Q4_K_M	128K+ context for full documents
Legal/compliance review	D — Maximum quality	Llama 3.3 70B Q8_0	Near-lossless for precision-critical work
Mixed diverse roles	B + occasional swap	GPT-oss-120B + 8B, swap to specialists	Covers everything

---

8. Models That Do NOT Fit at 96GB

No CPU offloading. No partial fits. Minimum Q4 quantization.

Model	Params	Architecture	Smallest Viable Quant	Weight Size	Verdict
DeepSeek-R1	671B	MoE	Q2_K	~250 GB	❌ Datacenter only (8× H100)
DeepSeek V3.2	685B	MoE	Q4_K_M	~386 GB	❌ Datacenter only
GLM-5	744B	MoE	Q4_K_M	~400+ GB	❌ Datacenter only
Kimi K2.5	1T	MoE	Q4_K_M	~621 GB	❌ Datacenter only
MiniMax M2.5	230B	MoE	Q4_K_M	~140 GB	❌ 44 GB over limit
Qwen3-235B-A22B	235B	MoE	Q4_K_M	~140+ GB	❌ Far too large
Llama 4 Maverick	402B	MoE	Q4_K_M	~240+ GB	❌ Far too large
Mixtral 8x22B	141B	MoE	Q4_K_M	~85.6 GB	❌ Fits weights but only ~10 GB KV = unusable
Qwen2.5-72B	72.7B	Dense	Q8_0	~73+ GB	⚠️ Tight — limited context. Llama 3.3 70B Q8_0 preferred
Llama 3.3 70B	70B	Dense	BF16	~140 GB	❌ Full precision doesn't fit

The models that remain out of reach at 96GB are truly massive: 200B+ MoE models and full-precision 70B+ dense models. These require either datacenter GPUs (A100 80GB, H100, B200) or multi-card 96GB setups — a configuration reserved for the multi-GPU consumer tier report.

---

9. Diminishing Returns Assessment — Is 96GB Worth It?

9.1 The Honest Calculus

From 48GB → 64GB (covered in Part 6), the primary gains were quantization upgrades (70B at Q5_K_M, 49B at Q8_0) and multi-model serving. The jump was modest.

From 64GB → 96GB, the story changes because of two models that didn't exist at the 64GB tier:

1. **GPT-oss-120B** — a genuine 120B-class model running at 120+ t/s. No amount of quantization trickery brings this model into 64GB. It's a category jump.

2. **Qwen3-Next-80B-A3B** — while the Q4_K_M (48.5 GB) technically fits at 64GB, the KV cache room on a 64GB setup (multi-GPU pipeline split) is severely constrained and the speed penalty from splitting a 48.5 GB MoE model is wasteful. On the native 96GB card, it has 47.5 GB of KV headroom for 128K+ context at full single-card speed.

Without these two models, 96GB would be a marginal upgrade over 64GB. With them, it represents a genuine capability tier for agent deployment.

9.2 Cost-Benefit

Configuration	Price	Best Model	Speed	Context
2× RTX 5090 (64GB)	~$4,000	Llama 3.3 70B Q5_K_M	~8–10 t/s (pipeline)	16–32K
RTX PRO 6000 (96GB)	~$8,000	GPT-oss-120B MXFP4	**120–163 t/s**	64–131K

The speed differential is staggering: 12–20× faster for a model that's also substantially stronger. The cost-per-useful-token-per-second strongly favors the native 96GB card over multi-GPU 64GB setups.

Versus cloud APIs: At ~$8,000 one-time cost, the RTX PRO 6000 running GPT-oss-120B or Qwen3-Next-80B-A3B pays for itself versus cloud API usage within 2–6 months for heavy agent workloads (assuming $100–300/month in API costs). The breakeven is faster for teams running 24/7 agent pipelines.

9.3 What 96GB Still Cannot Do

- **Cloud APIs still lead on the hardest tasks.** Claude Sonnet 4, GPT-5, Gemini 2.5 Pro, and o3 outperform any local 96GB model on the most complex multi-step reasoning, creative writing, and nuanced judgment calls.
- **The 200B+ MoE frontier is out of reach.** MiniMax M2.5 (230B, 80.2% SWE-bench), Qwen3-235B-A22B, and full DeepSeek-R1 require datacenter-class hardware.
- **Legal, medical, and compliance roles still benefit from cloud verification.** While 96GB models are substantially better than 48GB models for these domains, the liability risk on safety-critical decisions means human + cloud API review remains advisable.
- **Multimodal video understanding is limited.** Llama 4 Scout offers basic vision but no video processing. Production-quality multimodal remains a cloud API strength.

---

10. Conclusion & Recommended OpenClaw Model Stack at 96GB

10.1 Primary Recommendation

Priority	Model	Roles Covered	Key Advantage
Frontier workhorse	GPT-oss-120B MXFP4	11 roles (T3–T5): orchestrator, coding, debugging, security, SRE, planning, architecture, critic, QA/test, task planning	120+ t/s, 90% MMLU, configurable reasoning, native tool calling
Context king	Qwen3-Next-80B-A3B Q4_K_M (Instruct)	5 roles (T2–T3): research, content writing, document summary, RAG, market research	128K+ context at MoE speed, Arena-Hard 82.7
Reasoning champion	Qwen3-Next-80B-A3B Q4_K_M (Thinking)	3 roles (T4): math, STEM, algorithms	AIME 87.8%, GPQA 77.2%, LiveCodeBench 68.7%
Near-lossless dense	Llama 3.3 70B Q8_0	2 roles (T3–T5): fact-checking, legal review	Q8_0 = +0.001 perplexity, 32–48K context
Analytical specialist	Nemotron-Super-49B Q8_0	2 roles (T5): medical, financial	Reasoning toggle + tool calling at Q8_0
Speed + context	Qwen3-30B-A3B Q8_0 (MoE)	2 roles (T2–T3): synthesis, meetings	150+ t/s, 160K+ context
Quality general	Qwen3-32B Q8_0	6 roles (T2–T3): editing, email, DevOps, curriculum, landing pages, support	Near-lossless quality
Mid-range	Qwen3-14B Q4_K_M	8 roles (T1–T2): social, news, finance, SEO, travel, recipes	Fast single-GPU
Fast lightweight	Qwen3-8B Q4_K_M	8 roles (T1): routing, validation, alerts	186 t/s, co-loads with any model
Heartbeat	Qwen3-0.6B Q8_0	1 role (T1): health monitor	500+ t/s

10.2 Core Insight: MoE-First at Unprecedented Scale

The defining advantage of 96GB is running MoE models large enough to compete with frontier APIs. Both GPT-oss-120B (5.1B active of 117B) and Qwen3-Next-80B-A3B (3B active of 80B) deliver their quality through massive parameter pools with tiny activation footprints. This is why they're fast — token generation speed is primarily limited by active parameters, not total VRAM consumption.

The result is an unusual performance profile: **models that use 65–85 GB of VRAM but generate tokens at speeds comparable to 8B models.** This makes them practical as primary workhorse models for interactive agent pipelines in a way that dense 70B models (which generate at 25–30 t/s) are not.

10.3 Minimum Viable Stack

For the simplest effective OpenClaw deployment at 96GB:

1. **GPT-oss-120B MXFP4** (~65 GB) + **Qwen3-8B Q4_K_M** (~5.5 GB) simultaneously loaded — covers orchestrator, all coding, all Tier 1 routing, and most Tier 3–5 roles at zero swap latency
2. Swap to **Qwen3-Next-80B-A3B-Thinking Q4_K_M** (48.5 GB) for deep reasoning tasks (AIME 87.8%)
3. Swap to **Llama 3.3 70B Q8_0** (~60 GB) for precision-critical legal/compliance work

This three-model rotation covers all 57 roles. The GPT-oss-120B + Qwen3-8B dual-load handles 80%+ of tasks without any model swapping.

10.4 The 96GB Verdict

The RTX PRO 6000 Blackwell at 96GB is the first single-GPU configuration where local AI agent deployment stops being a compromise and becomes a genuine production-grade alternative to cloud APIs. The combination of GPT-oss-120B (frontier tool calling + reasoning at MoE speed) and Qwen3-Next-80B-A3B (frontier reasoning + ultra-long context at MoE speed) creates an agent stack that matches or approaches cloud API quality for the vast majority of agent roles — at 120+ tokens per second, with full data privacy, and zero per-token cost.

20 of 57 roles upgrade at 96GB. The remaining 37 were already well-served at lower tiers. The upgrades concentrate precisely where they matter most: orchestration, complex coding, deep reasoning, and long-context analysis. For teams building serious OpenClaw deployments where quality, privacy, and cost efficiency all matter, the 96GB tier is the sweet spot.